Areeb Ahmad

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LinkedIn | GitHub | Medium | Hugging Face | X

PROFESSIONAL SUMMARY

Applied AI Engineer with around 3.5 years of experience delivering end-to-end LLM-based systems for real-world business workflows. Specialised in agentic systems, retrieval-augmented generation (RAG) pipelines, document understanding, and processing unstructured data from PDFs, scanned documents, images, and mixed-format sources. Built production-ready AI solutions that integrate into existing applications via APIs, with a strong builder mindset and track record of shipping quickly in ambiguous environments. Experienced across AWS cloud deployment, FastAPI-based integrations, and communicating AI trade-offs clearly to non-technical stakeholders. Actively writes about applied AI and shares hands-on learnings through technical blogs on Medium, alongside open-source models and datasets published on Hugging Face.

EDUCATION

Bachelor of Software Engineering (2021-2025)
National University of Sciences and Technology (NUST) | Islamabad, Pakistan

TECHNICAL SKILLS

AI / LLM / Agents: LangGraph, LangChain, Multi-Agent Systems, RAG Pipelines, Retrieval-Augmented Generation, Structured LLM Output, LLM Fine-tuning (QLoRA/SFT/PEFT), Prompt Engineering, Multi-Provider LLM Routing (OpenAI / Gemini / Grok / Claude via OpenRouter), LangSmith, OpenAI API, Llama Index

Document & Data Processing: PDF parsing, OCR (Google Vision), unstructured data extraction, JSONB, Apache Airflow ETL

Search & Retrieval: pgvector (HNSW), OpenSearch, Elasticsearch, BM25, Cohere Reranker, RRF Fusion, RAGAS & DeepEval evaluation

Cloud & Deployment: AWS (Lambda, S3, EC2, MWAA, CloudWatch), Docker, Serverless Framework, RunPod, GitHub Actions CI/CD

Backend: FastAPI, Nodejs, Celery, PostgreSQL, SQLModel, Redis, REST APIs, WebSockets, Oracle Apex

Computer Vision: OpenCV, YOLO (fine-tuning), image processing pipelines

PROFESSIONAL EXPERIENCE

GoSaaS - AI Engineer | gosaas.ai
Remote *Jun 2025 – Present*
Oracle systems integrator delivering PLM, SCM, and ERP solutions to high-tech manufacturing and life sciences clients.

- Automated extraction from scanned invoices and industrial P&ID diagrams with computer vision pipelines (OpenCV, fine-tuned YOLO), and delivered image-based semantic search for enterprise client Cohu matching submitted parts images against CAD drawings, replacing manual extraction and lookup
- Enabled non-technical staff to query fragmented Oracle databases in plain language via knowledge graph agents, fine-tuned NL2SQL LLMs, and hybrid (semantic + keyword) OpenSearch/Elasticsearch pipelines, improving retrieval accuracy while cutting compute usage and ad-hoc engineering requests
- Eliminated a manual claims-handling workflow for enterprise client Shark Ninja by building an automated pipeline to process and settle overpayment claims
- Developed and submitted 8+ production-ready ERP and CRM agents with evaluation and monitoring capabilities using Oracle AI Agent Studio for the Oracle Marketplace, which are currently used by Oracle Fusion consultants to reduce manual labor.

Heartstamp – GTM Founding AI Engineer | heartstamp.com
Remote *Sep 2025 – Jul 2026*
AI greeting card platform generating personalised digital 3D and print-on-demand cards.

- Architected and deployed the product's core conversational agent: a multi-agent LangGraph system (StateGraph, checkpointed memory) routing across support, card-design, and creative agents, with a multi-turn layer (Pydantic structured outputs, mid-turn schema swapping, rolling-window context, image-intent gating) cutting broken conversations and wasted image generation

- Cut generation costs and enabled per-task model swaps and A/B testing without code changes by building the agent tool layer (card search, font retrieval, text-to-image / image-to-image / restyle) on provider-agnostic routing (OpenAI, Gemini, Grok, Claude) with Flagsmith-gated rollout
- Created and managed end to end AI Agent's monitoring and evaluation production pipeline from idea to MVP to real users to sales.
- Improved search relevance and app responsiveness under load via the card semantic search (PostgreSQL full-text, pgvector HNSW, RRF fusion, affinity re-ranking) and async backend (Celery/Redis, WebSocket streaming, FastAPI + pgvector), with DeepEval, LangSmith, and OpenTelemetry/Jaeger catching regressions pre-release

Codeacme - AI Engineer (Contract) | codeacme.com
Islamabad, Pakistan (Remote) *Mar 2025 – Sep 2025*
AI solutions company delivering production ML systems for global clients.

- Shortened deployment cycles and cut infra cost by building end-to-end MLOps pipelines from training through serverless deployment, removing always-on compute
- Increased project delivery rate by 20% by training and deploying fine-tuned diffusion models (LoRA/PEFT), cutting per-project turnaround
- Turned loosely defined client requirements into working prototypes quickly, iterating them into production-ready deliverables

Pam AI - AI Support Engineer (Contract) | pam.ai
Virginia, USA (Remote) *Dec 2024 – May 2025*
Voice AI platform for car dealerships, automating inbound calls, texts, and service booking.

- Reduced hallucinations in a production AI system by developing agentic pipelines with multi-tool calling, cutting bad responses that previously required human support escalation
- Deployed AWS Lambda functions via Serverless Framework with CloudWatch monitoring, scaling with variable load and surfacing production failures faster

Freelance AI Engineer
Remote *Oct 2024 – Mar 2025*

- Replaced manual document review with an end-to-end RAG pipeline for scanned tax PDFs (Google Vision OCR, Llama Index, fine-tuned BGE-small-en embeddings for +4.3% retrieval boost, hybrid BM25 + dense retrieval)
- Built a real-time RAG research pipeline (ArXiv ingestion, semantic chunking, fine-tuned embeddings, BM25 + ensemble retrieval, Cohere reranker) with RAGAS evaluation to tune retrieval quality
- Secured sensitive banking data on a self-hosted model with NUST BankLlama: LLaMA 3.2-3B fine-tuned via QLoRA, Qdrant + BM25 hybrid retrieval, and Guardrails for PII detection and jailbreak prevention

Tranzbox - Data Scientist (Quant) | tranzbox.au
Melbourne, Australia (Remote) *Jun 2023 – Sep 2024*
Fintech company building ML-powered investment and trading systems.

- Engineered data and feature pipelines over large-scale stock-market data to build training datasets for predictive trading models
- Raised market prediction accuracy by benchmarking classical ML (SVM, decision trees, random forests) and reinforcement learning (A3C), identifying which techniques held up on historical data
- Automated a previously manual workflow with Apache Airflow ETL pipelines (DAGs) for S3 extraction, model training, and deployment to AWS MWAA at scale
- Built CI/CD with GitHub Actions deploying ML models to S3 and AWS Lambda, making model releases repeatable and hands-off
- Led team of 4 engineers, delivering seamless ML platform module integration